Stage 5 Engineering Technology 200-hour Master

Engineering Technology 7–10 (2024) | 200 periods | New syllabus for 2027 | v2.0

Duration	200 × 60-minute periods	**Course**	Stage 5
Balance	40 theory | 40 folio | 120 practical	**Practical**	60% of course
Syllabus	Engineering Technology 7–10 (2024)	**Implementation**	From 2027
Status	Faculty-ready network program	**Local controls**	Teacher to confirm before use

Official authority: **Engineering Technology 7–10 Syllabus (2024)** | **Open the Engineering course site**

Course requirement: all four focus areas, at least four projects and at least one collaborative activity. Practical experiences are the majority of this program.

Exact Stage 5 outcomes

Outcome	Exact official wording
EGT5-COM-01	communicates ideas, concepts and solutions for engineering practice
EGT5-ENV-01	analyses relationships between engineering design, production and sustainability
EGT5-EVL-01	investigates and evaluates engineering systems and solutions
EGT5-GRP-01	develops and applies technical graphics in engineering contexts
EGT5-IVT-01	explains engineering practices and the influence of technologies used in engineering industries
EGT5-MEA-01	applies mechanical analysis and practical testing to investigate engineering concepts
EGT5-SAF-01	applies risk management and safe work practices in engineering contexts
EGT5-USE-01	selects and applies materials for engineering projects

Complete course coverage map

Focus area	Exact dot points	Scheduled projects	Primary periods
Structures	37	Water Tower; Concrete Beam	37
Mechanisms	43	Hydraulic Digger; Geared Lolly Dispenser	43
Control systems	50	Garden Watering System	50
Engineering specialisation	33	Wind-powered USB Charger	33

Teacher to confirm before use

ID	Confirm	Owner
TTC-ENG-01	Confirm the local timetable, project order, room allocation, class supports and approved adjustments.	Teacher / faculty
TTC-ENG-02	Confirm current assessment authority, task wording, weights, dates, conditions and submission methods.	Teacher / faculty
TTC-ENG-03	Confirm each practical step, materials, tools, machines, equipment, demonstrations, supervision, SOPs, PPE and current risk controls.	Teacher / faculty
TTC-ENG-04	Confirm collaborative testing groups, student roles, equitable participation and the evidence record.	Teacher / faculty

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 1–5

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
1	**Site/theory**	Water Tower | Engineering principles	Teach and unpack this syllabus learning in the Water Tower context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Water Tower learning module 1** **Open the Water Tower project page**	**Describe the function and purpose of structures , including how they provide access and shelter** **Official content** **EGT5-IVT-01**	Module response showing: Describe the function and purpose of structures , including how they provide access and shelter EGT5-IVT-01
2	**Project/folio**	Water Tower | Engineering principles	Students apply the syllabus learning to their Water Tower folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Water Tower folio** **Open the Water Tower project page**	**Identify functional engineering components that provide structural integrity, such as foundations, beams and…** **Official content** **EGT5-USE-01, EGT5-COM-01**	Folio evidence demonstrating: Identify functional engineering components that provide structural integrity, such as foundations, beams and bracing EGT5-USE-01, EGT5-COM-01
3	**Practical**	Water Tower | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Water Tower making, investigation or testing step and record authentic progress. **Open the Water Tower project page**	**Outline regulations, codes of practice and standards for an identified structure in Australia** **Official content** **EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Outline regulations, codes of practice and standards for an identified structure in Australia EGT5-SAF-01, EGT5-USE-01
4	**Practical**	Water Tower | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Water Tower making, investigation or testing step and record authentic progress. **Open the Water Tower project page**	**Apply scheduling, resource allocation and budgeting to plan and manage a structural engineering project** **Official content** **EGT5-ENV-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Apply scheduling, resource allocation and budgeting to plan and manage a structural engineering project EGT5-ENV-01, EGT5-SAF-01, EGT5-USE-01
5	**Practical**	Water Tower | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Water Tower making, investigation or testing step and record authentic progress. **Open the Water Tower project page**	Outline the importance of Indigenous Cultural and Intellectual Property (ICIP) of Aboriginal and/or Torres… **Official content** **EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Outline the importance of Indigenous Cultural and Intellectual Property (ICIP) of Aboriginal and/or Torres Strait Islander Peoples and how it is related to structures EGT5-SAF-01, EGT5-USE-01

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 6–10

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
6	**Site/theory**	Water Tower | Engineering principles	Teach and unpack this syllabus learning in the Water Tower context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Water Tower learning module 1** **Open the Water Tower project page**	**Outline the importance of Indigenous Cultural and Intellectual Property (ICIP) of Aboriginal and/or Torres… Official content** **EGT5-IVT-01**	Module response showing: Outline the importance of Indigenous Cultural and Intellectual Property (ICIP) of Aboriginal and/or Torres Strait Islander Peoples and how it is related to structures EGT5-IVT-01
7	**Project/folio**	Water Tower | Engineering principles	Students apply the syllabus learning to their Water Tower folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Water Tower folio** **Open the Water Tower project page**	**Investigate and identify the impact of a range of structures on the physical environment in Australia Official content** **EGT5-ENV-01, EGT5-COM-01**	Folio evidence demonstrating: Investigate and identify the impact of a range of structures on the physical environment in Australia EGT5-ENV-01, EGT5-COM-01
8	**Practical**	Water Tower | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Water Tower making, investigation or testing step and record authentic progress. **Open the Water Tower project page**	**Identify and apply engineering principles and processes, including types of forces and their effects… Official content** **EGT5-SAF-01, EGT5-USE-01, EGT5-MEA-01**	Teacher observation and project/test record demonstrating: Identify and apply engineering principles and processes, including types of forces and their effects, structural integrity and safety, to produce a functioning structure EGT5-SAF-01, EGT5-USE-01, EGT5-MEA-01
9	**Practical**	Water Tower | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Water Tower making, investigation or testing step and record authentic progress. **Open the Water Tower project page**	Discuss technical and enterprise skills used by structural engineers **Official content** **EGT5-IVT-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Discuss technical and enterprise skills used by structural engineers EGT5-IVT-01, EGT5-SAF-01, EGT5-USE-01
10	**Practical**	Water Tower | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Water Tower making, investigation or testing step and record authentic progress. **Open the Water Tower project page**	**Identify and implement safe work practices when producing a functioning structure Official content** **EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Identify and implement safe work practices when producing a functioning structure EGT5-SAF-01, EGT5-USE-01

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 11–15

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
11	**Site/theory**	Water Tower | Engineering principles	Teach and unpack this syllabus learning in the Water Tower context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Water Tower learning module 2** **Open the Water Tower project page**	**Discuss technical and enterprise skills used by structural engineers** **Official content** **EGT5-IVT-01**	Module response showing: Discuss technical and enterprise skills used by structural engineers EGT5-IVT-01
12	**Project/folio**	Water Tower | Engineering principles	Students apply the syllabus learning to their Water Tower folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Water Tower folio** **Open the Water Tower project page**	**Discuss sustainability and environmental considerations in structural or civil construction** **Official content** **EGT5-ENV-01, EGT5-USE-01, EGT5-COM-01**	Folio evidence demonstrating: Discuss sustainability and environmental considerations in structural or civil construction EGT5-ENV-01, EGT5-USE-01, EGT5-COM-01
13	**Practical**	Water Tower | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Water Tower making, investigation or testing step and record authentic progress. **Open the Water Tower project page**	Apply safe work practices throughout the design, production and testing of models and projects for a structure **Official content** **EGT5-SAF-01, EGT5-EVL-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Apply safe work practices throughout the design, production and testing of models and projects for a structure EGT5-SAF-01, EGT5-EVL-01, EGT5-USE-01
14	**Practical**	Water Tower | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Water Tower making, investigation or testing step and record authentic progress. **Open the Water Tower project page**	**Use spreadsheets to calculate material quantities and monitor project costs** **Official content** **EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Use spreadsheets to calculate material quantities and monitor project costs EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01
15	**Practical**	Water Tower | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Water Tower making, investigation or testing step and record authentic progress. **Open the Water Tower project page**	**Investigate engineered structures used by Aboriginal and/or Torres Strait Islander Peoples, such as shelters…** **Official content** **EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Investigate engineered structures used by Aboriginal and/or Torres Strait Islander Peoples, such as shelters and fish traps EGT5-SAF-01, EGT5-USE-01

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 16–20

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
16	**Site/theory**	Water Tower | Materials	Teach and unpack this syllabus learning in the Water Tower context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Water Tower learning module 3** **Open the Water Tower project page**	**Apply safe work practices throughout the design, production and testing of models and projects for a structure Official content** **EGT5-SAF-01, EGT5-EVL-01, EGT5-IVT-01**	Module response showing: Apply safe work practices throughout the design, production and testing of models and projects for a structure EGT5-SAF-01, EGT5-EVL-01, EGT5-IVT-01
17	**Project/folio**	Water Tower | Materials	Students apply the syllabus learning to their Water Tower folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Water Tower folio** **Open the Water Tower project page**	Identify how materials can be modified to improve properties, such as composite materials, laminates, heat… **Official content** **EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01**	Folio evidence demonstrating: Identify how materials can be modified to improve properties, such as composite materials, laminates, heat treatment and reinforcement EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01
18	**Practical**	Water Tower | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Water Tower making, investigation or testing step and record authentic progress. **Open the Water Tower project page**	**Classify engineering materials used in structures as pure metals, alloys, polymers and timbers Official content** **EGT5-USE-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Classify engineering materials used in structures as pure metals, alloys, polymers and timbers EGT5-USE-01, EGT5-SAF-01
19	**Practical**	Water Tower | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Water Tower making, investigation or testing step and record authentic progress. **Open the Water Tower project page**	**Identify the properties of a material, such as hardness, ductility, malleability, and tensile and compressive… Official content** **EGT5-USE-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Identify the properties of a material, such as hardness, ductility, malleability, and tensile and compressive strength, that make the materials suitable for use in structures EGT5-USE-01, EGT5-SAF-01
20	**Practical**	Water Tower | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Water Tower making, investigation or testing step and record authentic progress. **Open the Water Tower project page**	**Compare traditional and modern construction materials and describe their properties, such as timber versus… Official content** **EGT5-USE-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Compare traditional and modern construction materials and describe their properties, such as timber versus laminates and rammed earth versus concrete EGT5-USE-01, EGT5-SAF-01

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 21–25

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
21	**Site/theory**	Water Tower | Engineering principles	Teach and unpack this syllabus learning in the Water Tower context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Water Tower learning module 3** **Open the Water Tower project page**	Identify functional engineering components that provide structural integrity, such as foundations, beams and… **Official content** **EGT5-USE-01, EGT5-IVT-01**	Module response showing: Identify functional engineering components that provide structural integrity, such as foundations, beams and bracing EGT5-USE-01, EGT5-IVT-01
22	**Project/folio**	Water Tower | Materials	Students apply the syllabus learning to their Water Tower folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Water Tower folio** **Open the Water Tower project page**	**Identify how materials can be modified to improve properties, such as composite materials, laminates, heat… Official content** **EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01**	Folio evidence demonstrating: Identify how materials can be modified to improve properties, such as composite materials, laminates, heat treatment and reinforcement EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01
23	**Practical**	Water Tower | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Water Tower making, investigation or testing step and record authentic progress. **Open the Water Tower project page**	**Compare renewable and non-renewable resources and explain their advantages and limitations in the design and… Official content** **EGT5-ENV-01, EGT5-USE-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Compare renewable and non-renewable resources and explain their advantages and limitations in the design and construction of structures EGT5-ENV-01, EGT5-USE-01, EGT5-SAF-01
24	**Practical**	Water Tower | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Water Tower making, investigation or testing step and record authentic progress. **Open the Water Tower project page**	**Test a range of materials to determine the properties of hardness, corrosion resistance and malleability, and… Official content** **EGT5-USE-01, EGT5-EVL-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Test a range of materials to determine the properties of hardness, corrosion resistance and malleability, and record collected data EGT5-USE-01, EGT5-EVL-01, EGT5-SAF-01
25	**Practical**	Water Tower | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Water Tower making, investigation or testing step and record authentic progress. **Open the Water Tower project page**	Investigate and identify the impact of a range of structures on the physical environment in Australia **Official content** **EGT5-ENV-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Investigate and identify the impact of a range of structures on the physical environment in Australia EGT5-ENV-01, EGT5-SAF-01, EGT5-USE-01

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 26–30

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
26	**Site/theory**	Concrete Beam | Materials	Teach and unpack this syllabus learning in the Concrete Beam context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Concrete Beam learning module 1** **Open the Concrete Beam project page**	**Select and test the suitability of materials by constructing a project Official content EGT5-USE-01, EGT5-EVL-01, EGT5-IVT-01**	Module response showing: Select and test the suitability of materials by constructing a project EGT5-USE-01, EGT5-EVL-01, EGT5-IVT-01
27	**Project/folio**	Concrete Beam | Materials	Students apply the syllabus learning to their Concrete Beam folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Concrete Beam folio** **Open the Concrete Beam project page**	**Investigate engineering materials used by Aboriginal and/or Torres Strait Islander Peoples, such as timber… Official content EGT5-USE-01, EGT5-COM-01**	Folio evidence demonstrating: Investigate engineering materials used by Aboriginal and/or Torres Strait Islander Peoples, such as timber, resin and fibres EGT5-USE-01, EGT5-COM-01
28	**Practical**	Concrete Beam | Structural analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Concrete Beam making, investigation or testing step and record authentic progress. **Open the Concrete Beam project page**	**Identify forces that act on structures, including live loads , dead loads and calculate weight force Official content EGT5-MEA-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Identify forces that act on structures, including live loads , dead loads and calculate weight force EGT5-MEA-01, EGT5-SAF-01, EGT5-USE-01
29	**Practical**	Concrete Beam | Structural analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Concrete Beam making, investigation or testing step and record authentic progress. **Open the Concrete Beam project page**	Identify compressive and tensile forces acting on members in an identified structure **Official content EGT5-MEA-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Identify compressive and tensile forces acting on members in an identified structure EGT5-MEA-01, EGT5-SAF-01, EGT5-USE-01
30	**Practical**	Concrete Beam | Structural analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Concrete Beam making, investigation or testing step and record authentic progress. **Open the Concrete Beam project page**	**Identify compressive and tensile forces acting on members in an identified structure Official content EGT5-MEA-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Identify compressive and tensile forces acting on members in an identified structure EGT5-MEA-01, EGT5-SAF-01, EGT5-USE-01

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 31–35

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
31	**Site/theory**	Concrete Beam | Structural analysis	Teach and unpack this syllabus learning in the Concrete Beam context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Concrete Beam learning module 1** **Open the Concrete Beam project page**	**Construct pin-jointed structures using different shapes to explore basic structural concepts, such as force…** **Official content** **EGT5-MEA-01, EGT5-IVT-01**	Module response showing: Construct pin-jointed structures using different shapes to explore basic structural concepts, such as force, load , reactions and equilibrium EGT5-MEA-01, EGT5-IVT-01
32	**Project/folio**	Concrete Beam | Structural analysis	Students apply the syllabus learning to their Concrete Beam folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Concrete Beam folio** **Open the Concrete Beam project page**	**Contrast destructive and non-destructive testing and how they are used to inform engineering decisions** **Official content** **EGT5-EVL-01, EGT5-COM-01**	Folio evidence demonstrating: Contrast destructive and non-destructive testing and how they are used to inform engineering decisions EGT5-EVL-01, EGT5-COM-01
33	**Practical**	Concrete Beam | Structural analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Concrete Beam making, investigation or testing step and record authentic progress. **Open the Concrete Beam project page**	Use the results of testing to modify pin-jointed structures **Official content** **EGT5-EVL-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Use the results of testing to modify pin-jointed structures EGT5-EVL-01, EGT5-SAF-01, EGT5-USE-01
34	**Practical**	Concrete Beam | Structural analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Concrete Beam making, investigation or testing step and record authentic progress. **Open the Concrete Beam project page**	**Test and document the effects of identified forces on pin-jointed structures** **Official content** **EGT5-EVL-01, EGT5-MEA-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Test and document the effects of identified forces on pin-jointed structures EGT5-EVL-01, EGT5-MEA-01, EGT5-SAF-01
35	**Practical**	Concrete Beam | Structural analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Concrete Beam making, investigation or testing step and record authentic progress. **Open the Concrete Beam project page**	**Use the results of testing to modify pin-jointed structures** **Official content** **EGT5-EVL-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Use the results of testing to modify pin-jointed structures EGT5-EVL-01, EGT5-SAF-01, EGT5-USE-01

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 36–40

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
36	**Site/theory**	Concrete Beam | Communication	Teach and unpack this syllabus learning in the Concrete Beam context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Concrete Beam learning module 2** **Open the Concrete Beam project page**	Identify key components of an engineering drawing, and the standards from AS 1100 relevant to the engineering… **Official content** **EGT5-SAF-01, EGT5-GRP-01, EGT5-IVT-01**	Module response showing: Identify key components of an engineering drawing, and the standards from AS 1100 relevant to the engineering profession, when drawing structures EGT5-SAF-01, EGT5-GRP-01, EGT5-IVT-01
37	**Project/folio**	Concrete Beam | Structural analysis	Students apply the syllabus learning to their Concrete Beam folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Concrete Beam folio** **Open the Concrete Beam project page**	**Build and test a working model of a structure for a specific purpose using appropriate tools and processes Official content** **EGT5-USE-01, EGT5-EVL-01, EGT5-COM-01**	Folio evidence demonstrating: Build and test a working model of a structure for a specific purpose using appropriate tools and processes EGT5-USE-01, EGT5-EVL-01, EGT5-COM-01
38	**Practical**	Concrete Beam | Structural analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Concrete Beam making, investigation or testing step and record authentic progress. **Open the Concrete Beam project page**	**Evaluate the use of structural elements in projects Official content** **EGT5-EVL-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Evaluate the use of structural elements in projects EGT5-EVL-01, EGT5-SAF-01, EGT5-USE-01
39	**Practical**	Concrete Beam | Communication	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Concrete Beam making, investigation or testing step and record authentic progress. **Open the Concrete Beam project page**	**Identify key components of an engineering drawing, and the standards from AS 1100 relevant to the engineering… Official content** **EGT5-SAF-01, EGT5-GRP-01, EGT5-IVT-01**	Teacher observation and project/test record demonstrating: Identify key components of an engineering drawing, and the standards from AS 1100 relevant to the engineering profession, when drawing structures EGT5-SAF-01, EGT5-GRP-01, EGT5-IVT-01
40	**Practical**	Concrete Beam | Communication	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Concrete Beam making, investigation or testing step and record authentic progress. **Open the Concrete Beam project page**	Document material selection and justification, structural analysis and test results in an engineering report **Official content** **EGT5-COM-01, EGT5-USE-01, EGT5-EVL-01**	Teacher observation and project/test record demonstrating: Document material selection and justification, structural analysis and test results in an engineering report EGT5-COM-01, EGT5-USE-01, EGT5-EVL-01

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 41–45

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
41	**Site/theory**	Concrete Beam | Communication	Teach and unpack this syllabus learning in the Concrete Beam context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Concrete Beam learning module 3** **Open the Concrete Beam project page**	**Produce a pictorial sketch of an identified structure, such as an isometric or oblique drawing Official content** **EGT5-GRP-01, EGT5-USE-01, EGT5-IVT-01**	Module response showing: Produce a pictorial sketch of an identified structure, such as an isometric or oblique drawing EGT5-GRP-01, EGT5-USE-01, EGT5-IVT-01
42	**Project/folio**	Concrete Beam | Communication	Students apply the syllabus learning to their Concrete Beam folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Concrete Beam folio** **Open the Concrete Beam project page**	**Develop an orthogonal engineering drawing of a structure, including 2 views and dimensions using third-angle… Official content** **EGT5-GRP-01, EGT5-COM-01**	Folio evidence demonstrating: Develop an orthogonal engineering drawing of a structure, including 2 views and dimensions using third-angle projection EGT5-GRP-01, EGT5-COM-01
43	**Practical**	Concrete Beam | Communication	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Concrete Beam making, investigation or testing step and record authentic progress. **Open the Concrete Beam project page**	Use subject-specific terminology to communicate concepts of engineering structures **Official content** **EGT5-COM-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Use subject-specific terminology to communicate concepts of engineering structures EGT5-COM-01, EGT5-SAF-01, EGT5-USE-01
44	**Practical**	Concrete Beam | Communication	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Concrete Beam making, investigation or testing step and record authentic progress. **Open the Concrete Beam project page**	**Produce a parts list to prepare materials for the construction of a structure Official content** **EGT5-GRP-01, EGT5-USE-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Produce a parts list to prepare materials for the construction of a structure EGT5-GRP-01, EGT5-USE-01, EGT5-SAF-01
45	**Practical**	Concrete Beam | Communication	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Concrete Beam making, investigation or testing step and record authentic progress. **Open the Concrete Beam project page**	**Document material selection and justification, structural analysis and test results in an engineering report Official content** **EGT5-COM-01, EGT5-USE-01, EGT5-EVL-01**	Teacher observation and project/test record demonstrating: Document material selection and justification, structural analysis and test results in an engineering report EGT5-COM-01, EGT5-USE-01, EGT5-EVL-01

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 46–50

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
46	**Site/theory**	Concrete Beam | Communication	Teach and unpack this syllabus learning in the Concrete Beam context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Concrete Beam learning module 3** **Open the Concrete Beam project page**	**Apply data presentation techniques when presenting structural test results Official content EGT5-COM-01, EGT5-EVL-01, EGT5-IVT-01**	Module response showing: Apply data presentation techniques when presenting structural test results EGT5-COM-01, EGT5-EVL-01, EGT5-IVT-01
47	**Project/folio**	Concrete Beam | Structural analysis	Students apply the syllabus learning to their Concrete Beam folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Concrete Beam folio** **Open the Concrete Beam project page**	Identify compressive and tensile forces acting on members in an identified structure **Official content** **EGT5-MEA-01, EGT5-COM-01**	Folio evidence demonstrating: Identify compressive and tensile forces acting on members in an identified structure EGT5-MEA-01, EGT5-COM-01
48	**Practical**	Concrete Beam | Communication	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Concrete Beam making, investigation or testing step and record authentic progress. **Open the Concrete Beam project page**	**Create written texts to explain and evaluate factors that influence the design and engineering of structures Official content EGT5-COM-01, EGT5-EVL-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Create written texts to explain and evaluate factors that influence the design and engineering of structures EGT5-COM-01, EGT5-EVL-01, EGT5-SAF-01
49	**Practical**	Concrete Beam | Communication	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Concrete Beam making, investigation or testing step and record authentic progress. **Open the Concrete Beam project page**	**Use subject-specific terminology to communicate concepts of engineering structures Official content EGT5-COM-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Use subject-specific terminology to communicate concepts of engineering structures EGT5-COM-01, EGT5-SAF-01, EGT5-USE-01
50	**Practical**	Concrete Beam | Structural analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Concrete Beam making, investigation or testing step and record authentic progress. **Open the Concrete Beam project page**	Test and document the effects of identified forces on pin-jointed structures **Official content** **EGT5-EVL-01, EGT5-MEA-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Test and document the effects of identified forces on pin-jointed structures EGT5-EVL-01, EGT5-MEA-01, EGT5-SAF-01

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 51–55

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
51	**Site/theory**	Hydraulic Digger | Engineering principles	Teach and unpack this syllabus learning in the Hydraulic Digger context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Hydraulic Digger learning module 1** **Open the Hydraulic Digger project page**	**Describe the function and purpose of mechanisms , including how they operate as part of a system or machine Official content** **EGT5-USE-01, EGT5-MEA-01, EGT5-IVT-01**	Module response showing: Describe the function and purpose of mechanisms , including how they operate as part of a system or machine EGT5-USE-01, EGT5-MEA-01, EGT5-IVT-01
52	**Project/folio**	Hydraulic Digger | Engineering principles	Students apply the syllabus learning to their Hydraulic Digger folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Hydraulic Digger folio** **Open the Hydraulic Digger project page**	**Identify the functional components of a mechanism, such as axles, levers, pulleys, springs, gears and screws Official content** **EGT5-USE-01, EGT5-MEA-01, EGT5-COM-01**	Folio evidence demonstrating: Identify the functional components of a mechanism, such as axles, levers, pulleys, springs, gears and screws EGT5-USE-01, EGT5-MEA-01, EGT5-COM-01
53	**Practical**	Hydraulic Digger | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Hydraulic Digger making, investigation or testing step and record authentic progress. **Open the Hydraulic Digger project page**	**Identify and use safe work practices when participating in testing and developing mechanical projects Official content** **EGT5-SAF-01, EGT5-EVL-01, EGT5-MEA-01**	Teacher observation and project/test record demonstrating: Identify and use safe work practices when participating in testing and developing mechanical projects EGT5-SAF-01, EGT5-EVL-01, EGT5-MEA-01
54	**Practical**	Hydraulic Digger | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Hydraulic Digger making, investigation or testing step and record authentic progress. **Open the Hydraulic Digger project page**	**Dismantle and reassemble a mechanism to outline its function Official content** **EGT5-MEA-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Dismantle and reassemble a mechanism to outline its function EGT5-MEA-01, EGT5-SAF-01, EGT5-USE-01
55	**Practical**	Hydraulic Digger | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Hydraulic Digger making, investigation or testing step and record authentic progress. **Open the Hydraulic Digger project page**	**Apply scheduling, resource allocation and budgeting to plan and manage a mechanical engineering project Official content** **EGT5-ENV-01, EGT5-MEA-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Apply scheduling, resource allocation and budgeting to plan and manage a mechanical engineering project EGT5-ENV-01, EGT5-MEA-01, EGT5-SAF-01

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 56–60

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
56	**Site/theory**	Hydraulic Digger | Engineering principles	Teach and unpack this syllabus learning in the Hydraulic Digger context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Hydraulic Digger learning module 1** **Open the Hydraulic Digger project page**	**Investigate how Aboriginal and/or Torres Strait Islander Peoples used levers and pulleys Official content** **EGT5-IVT-01**	Module response showing: Investigate how Aboriginal and/or Torres Strait Islander Peoples used levers and pulleys EGT5-IVT-01
57	**Project/folio**	Hydraulic Digger | Engineering principles	Students apply the syllabus learning to their Hydraulic Digger folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Hydraulic Digger folio** **Open the Hydraulic Digger project page**	**Test the mechanical advantage and efficiency of simple machines , including a lever, wheel and axle, pulley… Official content** **EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01**	Folio evidence demonstrating: Test the mechanical advantage and efficiency of simple machines , including a lever, wheel and axle, pulley, gear and spring EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01
58	**Practical**	Hydraulic Digger | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Hydraulic Digger making, investigation or testing step and record authentic progress. **Open the Hydraulic Digger project page**	**Describe key innovations based on simple machines to understand how mechanical engineering has evolved over time Official content** **EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Describe key innovations based on simple machines to understand how mechanical engineering has evolved over time EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01
59	**Practical**	Hydraulic Digger | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Hydraulic Digger making, investigation or testing step and record authentic progress. **Open the Hydraulic Digger project page**	Explain how simple machines work together in a mechanical system **Official content** **EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Explain how simple machines work together in a mechanical system EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01
60	**Practical**	Hydraulic Digger | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Hydraulic Digger making, investigation or testing step and record authentic progress. **Open the Hydraulic Digger project page**	**Explain how simple machines work together in a mechanical system Official content** **EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Explain how simple machines work together in a mechanical system EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 61–65

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
61	**Site/theory**	Hydraulic Digger | Engineering principles	Teach and unpack this syllabus learning in the Hydraulic Digger context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Hydraulic Digger learning module 2** **Open the Hydraulic Digger project page**	**Identify and apply linear, rotary, oscillating and reciprocating motion in various mechanical devices Official content** **EGT5-MEA-01, EGT5-IVT-01**	Module response showing: Identify and apply linear, rotary, oscillating and reciprocating motion in various mechanical devices EGT5-MEA-01, EGT5-IVT-01
62	**Project/folio**	Hydraulic Digger | Engineering principles	Students apply the syllabus learning to their Hydraulic Digger folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Hydraulic Digger folio** **Open the Hydraulic Digger project page**	**Use and modify existing designs when completing projects Official content** **EGT5-EVL-01, EGT5-COM-01**	Folio evidence demonstrating: Use and modify existing designs when completing projects EGT5-EVL-01, EGT5-COM-01
63	**Practical**	Hydraulic Digger | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Hydraulic Digger making, investigation or testing step and record authentic progress. **Open the Hydraulic Digger project page**	**Construct a model of a mechanism that converts circular motion to linear motion Official content** **EGT5-MEA-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Construct a model of a mechanism that converts circular motion to linear motion EGT5-MEA-01, EGT5-SAF-01, EGT5-USE-01
64	**Practical**	Hydraulic Digger | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Hydraulic Digger making, investigation or testing step and record authentic progress. **Open the Hydraulic Digger project page**	**Discuss the application of precision measuring tools, such as micrometers and vernier calipers, to ensure… Official content** **EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Discuss the application of precision measuring tools, such as micrometers and vernier calipers, to ensure accuracy in the production of mechanical parts EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01
65	**Practical**	Hydraulic Digger | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Hydraulic Digger making, investigation or testing step and record authentic progress. **Open the Hydraulic Digger project page**	**Calculate quantities and costs of materials and components used in the completion of mechanical projects Official content** **EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Calculate quantities and costs of materials and components used in the completion of mechanical projects EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 66–70

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
66	**Site/theory**	Hydraulic Digger | Engineering principles	Teach and unpack this syllabus learning in the Hydraulic Digger context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Hydraulic Digger learning module 3** **Open the Hydraulic Digger project page**	**Investigate the role of mechanical engineering in the development of robotics used in automation Official content** **EGT5-MEA-01, EGT5-IVT-01**	Module response showing: Investigate the role of mechanical engineering in the development of robotics used in automation EGT5-MEA-01, EGT5-IVT-01
67	**Project/folio**	Hydraulic Digger | Engineering principles	Students apply the syllabus learning to their Hydraulic Digger folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Hydraulic Digger folio** **Open the Hydraulic Digger project page**	Evaluate and use mechanical components to solve an identified problem in practical projects **Official content** **EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01**	Folio evidence demonstrating: Evaluate and use mechanical components to solve an identified problem in practical projects EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01
68	**Practical**	Hydraulic Digger | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Hydraulic Digger making, investigation or testing step and record authentic progress. **Open the Hydraulic Digger project page**	**Discuss sustainability considerations in mechanical engineering, such as product life cycle and reusability of… Official content** **EGT5-ENV-01, EGT5-USE-01, EGT5-MEA-01**	Teacher observation and project/test record demonstrating: Discuss sustainability considerations in mechanical engineering, such as product life cycle and reusability of machine parts EGT5-ENV-01, EGT5-USE-01, EGT5-MEA-01
69	**Practical**	Hydraulic Digger | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Hydraulic Digger making, investigation or testing step and record authentic progress. **Open the Hydraulic Digger project page**	**Evaluate and use mechanical components to solve an identified problem in practical projects Official content** **EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01**	Teacher observation and project/test record demonstrating: Evaluate and use mechanical components to solve an identified problem in practical projects EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01
70	**Practical**	Hydraulic Digger | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Hydraulic Digger making, investigation or testing step and record authentic progress. **Open the Hydraulic Digger project page**	**Apply safe work practices throughout the design, production and testing of models and projects for a mechanism Official content** **EGT5-SAF-01, EGT5-EVL-01, EGT5-MEA-01**	Teacher observation and project/test record demonstrating: Apply safe work practices throughout the design, production and testing of models and projects for a mechanism EGT5-SAF-01, EGT5-EVL-01, EGT5-MEA-01

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 71–75

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
71	**Site/theory**	Hydraulic Digger | Materials	Teach and unpack this syllabus learning in the Hydraulic Digger context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Hydraulic Digger learning module 3** **Open the Hydraulic Digger project page**	**Identify the properties of a material, including hardness, machineability and wear resistance, that make it… Official content** **EGT5-USE-01, EGT5-MEA-01, EGT5-IVT-01**	Module response showing: Identify the properties of a material, including hardness, machineability and wear resistance, that make it suitable for use in mechanisms EGT5-USE-01, EGT5-MEA-01, EGT5-IVT-01
72	**Project/folio**	Hydraulic Digger | Materials	Students apply the syllabus learning to their Hydraulic Digger folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Hydraulic Digger folio** **Open the Hydraulic Digger project page**	**Compare the general properties of traditional and modern materials used in manufacturing mechanisms, including… Official content** **EGT5-USE-01, EGT5-MEA-01, EGT5-COM-01**	Folio evidence demonstrating: Compare the general properties of traditional and modern materials used in manufacturing mechanisms, including metals and composites EGT5-USE-01, EGT5-MEA-01, EGT5-COM-01
73	**Practical**	Hydraulic Digger | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Hydraulic Digger making, investigation or testing step and record authentic progress. **Open the Hydraulic Digger project page**	**Apply work hardening and heat treatment processes to modify a range of materials, and test how these processes… Official content** **EGT5-USE-01, EGT5-EVL-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Apply work hardening and heat treatment processes to modify a range of materials, and test how these processes change the properties of the material EGT5-USE-01, EGT5-EVL-01, EGT5-SAF-01
74	**Practical**	Hydraulic Digger | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Hydraulic Digger making, investigation or testing step and record authentic progress. **Open the Hydraulic Digger project page**	**Use data to assess the properties of a range of materials to evaluate their performance under identified… Official content** **EGT5-ENV-01, EGT5-USE-01, EGT5-EVL-01**	Teacher observation and project/test record demonstrating: Use data to assess the properties of a range of materials to evaluate their performance under identified conditions, such as temperature extremes and corrosive environments EGT5-ENV-01, EGT5-USE-01, EGT5-EVL-01
75	**Practical**	Hydraulic Digger | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Hydraulic Digger making, investigation or testing step and record authentic progress. **Open the Hydraulic Digger project page**	Identify and use safe work practices when participating in testing and developing mechanical projects **Official content** **EGT5-SAF-01, EGT5-EVL-01, EGT5-MEA-01**	Teacher observation and project/test record demonstrating: Identify and use safe work practices when participating in testing and developing mechanical projects EGT5-SAF-01, EGT5-EVL-01, EGT5-MEA-01

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 76–80

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
76	**Site/theory**	Geared Lolly Dispenser | Materials	Teach and unpack this syllabus learning in the Geared Lolly Dispenser context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Geared Lolly Dispenser learning module 1** **Open the Geared Lolly Dispenser project page**	**Select and assess the suitability of materials used to construct a mechanism for an intended purpose** Official content **EGT5-USE-01, EGT5-MEA-01, EGT5-IVT-01**	Module response showing: Select and assess the suitability of materials used to construct a mechanism for an intended purpose EGT5-USE-01, EGT5-MEA-01, EGT5-IVT-01
77	**Project/folio**	Geared Lolly Dispenser | Materials	Students apply the syllabus learning to their Geared Lolly Dispenser folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Geared Lolly Dispenser evidence guide** **Open the Geared Lolly Dispenser project page**	**Investigate the effect of friction between materials, including how friction could be either an advantage or…** Official content **EGT5-USE-01, EGT5-MEA-01, EGT5-COM-01**	Folio evidence demonstrating: Investigate the effect of friction between materials, including how friction could be either an advantage or disadvantage in a mechanism EGT5-USE-01, EGT5-MEA-01, EGT5-COM-01
78	**Practical**	Geared Lolly Dispenser | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Geared Lolly Dispenser making, investigation or testing step and record authentic progress. **Open the Geared Lolly Dispenser project page**	**Test the factors that impact on the corrosion of a mechanical component** Official content **EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01**	Teacher observation and project/test record demonstrating: Test the factors that impact on the corrosion of a mechanical component EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01
79	**Practical**	Geared Lolly Dispenser | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Geared Lolly Dispenser making, investigation or testing step and record authentic progress. **Open the Geared Lolly Dispenser project page**	**Outline and apply a range of processes and techniques, such as the application of finishes, to protect…** Official content **EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Outline and apply a range of processes and techniques, such as the application of finishes, to protect mechanical components from corrosion EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01
80	**Practical**	Geared Lolly Dispenser | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Geared Lolly Dispenser making, investigation or testing step and record authentic progress. **Open the Geared Lolly Dispenser project page**	**Investigate the role of the waste hierarchy in material selection and use during the production of mechanisms** Official content **EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Investigate the role of the waste hierarchy in material selection and use during the production of mechanisms EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 81–85

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
81	**Site/theory**	Geared Lolly Dispenser | Mechanical analysis	Teach and unpack this syllabus learning in the Geared Lolly Dispenser context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Geared Lolly Dispenser learning module 1** **Open the Geared Lolly Dispenser project page**	**Test a range of simple mechanisms, including levers, pulleys and gears, to determine their efficiency** Official content **EGT5-EVL-01, EGT5-MEA-01, EGT5-IVT-01**	Module response showing: Test a range of simple mechanisms, including levers, pulleys and gears, to determine their efficiency EGT5-EVL-01, EGT5-MEA-01, EGT5-IVT-01
82	**Project/folio**	Geared Lolly Dispenser | Mechanical analysis	Students apply the syllabus learning to their Geared Lolly Dispenser folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Geared Lolly Dispenser evidence guide** **Open the Geared Lolly Dispenser project page**	Analyse the functional effect of force and the related output of simple mechanisms, such as levers, pulleys… **Official content** **EGT5-MEA-01, EGT5-COM-01**	Folio evidence demonstrating: Analyse the functional effect of force and the related output of simple mechanisms, such as levers, pulleys and gears EGT5-MEA-01, EGT5-COM-01
83	**Practical**	Geared Lolly Dispenser | Mechanical analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Geared Lolly Dispenser making, investigation or testing step and record authentic progress. **Open the Geared Lolly Dispenser project page**	**Analyse the functional effect of force and the related output of simple mechanisms, such as levers, pulleys…** Official content **EGT5-MEA-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Analyse the functional effect of force and the related output of simple mechanisms, such as levers, pulleys and gears EGT5-MEA-01, EGT5-SAF-01, EGT5-USE-01
84	**Practical**	Geared Lolly Dispenser | Mechanical analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Geared Lolly Dispenser making, investigation or testing step and record authentic progress. **Open the Geared Lolly Dispenser project page**	**Build and test a working model of a mechanism using appropriate tools and processes** Official content **EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01**	Teacher observation and project/test record demonstrating: Build and test a working model of a mechanism using appropriate tools and processes EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01
85	**Practical**	Geared Lolly Dispenser | Mechanical analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Geared Lolly Dispenser making, investigation or testing step and record authentic progress. **Open the Geared Lolly Dispenser project page**	**Compare the forces acting on a model of a mechanism and assess their efficiency** Official content **EGT5-MEA-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Compare the forces acting on a model of a mechanism and assess their efficiency EGT5-MEA-01, EGT5-SAF-01, EGT5-USE-01

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 86–90

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
86	**Site/theory**	Geared Lolly Dispenser | Mechanical analysis	Teach and unpack this syllabus learning in the Geared Lolly Dispenser context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Geared Lolly Dispenser learning module 2** **Open the Geared Lolly Dispenser project page**	**Test and evaluate how mechanical systems relate to and interact with other systems to demonstrate their…** **Official content** **EGT5-EVL-01, EGT5-MEA-01, EGT5-IVT-01**	Module response showing: Test and evaluate how mechanical systems relate to and interact with other systems to demonstrate their dependencies on each other EGT5-EVL-01, EGT5-MEA-01, EGT5-IVT-01
87	**Project/folio**	Geared Lolly Dispenser | Mechanical analysis	Students apply the syllabus learning to their Geared Lolly Dispenser folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Geared Lolly Dispenser evidence guide** **Open the Geared Lolly Dispenser project page**	**Apply an engineering process to produce a functioning mechanism based on mechanical principles** **Official content** **EGT5-USE-01, EGT5-MEA-01, EGT5-COM-01**	Folio evidence demonstrating: Apply an engineering process to produce a functioning mechanism based on mechanical principles EGT5-USE-01, EGT5-MEA-01, EGT5-COM-01
88	**Practical**	Geared Lolly Dispenser | Mechanical analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Geared Lolly Dispenser making, investigation or testing step and record authentic progress. **Open the Geared Lolly Dispenser project page**	Evaluate the use of mechanical elements in projects **Official content** **EGT5-EVL-01, EGT5-MEA-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Evaluate the use of mechanical elements in projects EGT5-EVL-01, EGT5-MEA-01, EGT5-SAF-01
89	**Practical**	Geared Lolly Dispenser | Mechanical analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Geared Lolly Dispenser making, investigation or testing step and record authentic progress. **Open the Geared Lolly Dispenser project page**	**Identify and explain the concept of mechanical advantage (MA) in a simple mechanical system** **Official content** **EGT5-MEA-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Identify and explain the concept of mechanical advantage (MA) in a simple mechanical system EGT5-MEA-01, EGT5-SAF-01, EGT5-USE-01
90	**Practical**	Geared Lolly Dispenser | Mechanical analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Geared Lolly Dispenser making, investigation or testing step and record authentic progress. **Open the Geared Lolly Dispenser project page**	**Evaluate the use of mechanical elements in projects** **Official content** **EGT5-EVL-01, EGT5-MEA-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Evaluate the use of mechanical elements in projects EGT5-EVL-01, EGT5-MEA-01, EGT5-SAF-01

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 91–95

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
91	**Site/theory**	Geared Lolly Dispenser | Communication	Teach and unpack this syllabus learning in the Geared Lolly Dispenser context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Geared Lolly Dispenser learning module 3** **Open the Geared Lolly Dispenser project page**	**Develop an engineering drawing using the AS 1100 for drawing mechanisms** **Official content** **EGT5-GRP-01, EGT5-MEA-01, EGT5-IVT-01**	Module response showing: Develop an engineering drawing using the AS 1100 for drawing mechanisms EGT5-GRP-01, EGT5-MEA-01, EGT5-IVT-01
92	**Project/folio**	Geared Lolly Dispenser | Communication	Students apply the syllabus learning to their Geared Lolly Dispenser folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Geared Lolly Dispenser evidence guide** **Open the Geared Lolly Dispenser project page**	**Produce a pictorial drawing of a simple mechanical component, such as an isometric or oblique drawing** **Official content** **EGT5-GRP-01, EGT5-USE-01, EGT5-MEA-01**	Folio evidence demonstrating: Produce a pictorial drawing of a simple mechanical component, such as an isometric or oblique drawing EGT5-GRP-01, EGT5-USE-01, EGT5-MEA-01
93	**Practical**	Geared Lolly Dispenser | Communication	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Geared Lolly Dispenser making, investigation or testing step and record authentic progress. **Open the Geared Lolly Dispenser project page**	**Develop an orthogonal engineering drawing of a mechanical component, including 2 views and dimensions using…** **Official content** **EGT5-GRP-01, EGT5-USE-01, EGT5-MEA-01**	Teacher observation and project/test record demonstrating: Develop an orthogonal engineering drawing of a mechanical component, including 2 views and dimensions using third-angle projection EGT5-GRP-01, EGT5-USE-01, EGT5-MEA-01
94	**Practical**	Geared Lolly Dispenser | Communication	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Geared Lolly Dispenser making, investigation or testing step and record authentic progress. **Open the Geared Lolly Dispenser project page**	Document material selection and justification, mechanical analysis and test results in an engineering report **Official content** **EGT5-COM-01, EGT5-USE-01, EGT5-EVL-01**	Teacher observation and project/test record demonstrating: Document material selection and justification, mechanical analysis and test results in an engineering report EGT5-COM-01, EGT5-USE-01, EGT5-EVL-01
95	**Practical**	Geared Lolly Dispenser | Communication	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Geared Lolly Dispenser making, investigation or testing step and record authentic progress. **Open the Geared Lolly Dispenser project page**	**Apply a sequence of skills to develop an engineering drawing using computer-aided design (CAD) software** **Official content** **EGT5-GRP-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Apply a sequence of skills to develop an engineering drawing using computer-aided design (CAD) software EGT5-GRP-01, EGT5-SAF-01, EGT5-USE-01

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 96–100

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
96	**Site/theory**	Geared Lolly Dispenser | Communication	Teach and unpack this syllabus learning in the Geared Lolly Dispenser context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Geared Lolly Dispenser learning module 3** **Open the Geared Lolly Dispenser project page**	**Produce a parts list to prepare materials for the construction of a mechanism** Official content **EGT5-GRP-01, EGT5-USE-01, EGT5-MEA-01**	Module response showing: Produce a parts list to prepare materials for the construction of a mechanism EGT5-GRP-01, EGT5-USE-01, EGT5-MEA-01
97	**Project/folio**	Geared Lolly Dispenser | Communication	Students apply the syllabus learning to their Geared Lolly Dispenser folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Geared Lolly Dispenser evidence guide** **Open the Geared Lolly Dispenser project page**	**Document material selection and justification, mechanical analysis and test results in an engineering report** Official content **EGT5-COM-01, EGT5-USE-01, EGT5-EVL-01**	Folio evidence demonstrating: Document material selection and justification, mechanical analysis and test results in an engineering report EGT5-COM-01, EGT5-USE-01, EGT5-EVL-01
98	**Practical**	Geared Lolly Dispenser | Communication	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Geared Lolly Dispenser making, investigation or testing step and record authentic progress. **Open the Geared Lolly Dispenser project page**	**Create written texts to explain and justify decision-making in the development of a solution** Official content **EGT5-COM-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Create written texts to explain and justify decision-making in the development of a solution EGT5-COM-01, EGT5-SAF-01, EGT5-USE-01
99	**Practical**	Geared Lolly Dispenser | Communication	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Geared Lolly Dispenser making, investigation or testing step and record authentic progress. **Open the Geared Lolly Dispenser project page**	**Use subject-specific terminology to communicate concepts of engineering mechanisms** Official content **EGT5-COM-01, EGT5-MEA-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Use subject-specific terminology to communicate concepts of engineering mechanisms EGT5-COM-01, EGT5-MEA-01, EGT5-SAF-01
100	**Practical**	Geared Lolly Dispenser | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Geared Lolly Dispenser making, investigation or testing step and record authentic progress. **Open the Geared Lolly Dispenser project page**	Outline and apply a range of processes and techniques, such as the application of finishes, to protect… **Official content** **EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Outline and apply a range of processes and techniques, such as the application of finishes, to protect mechanical components from corrosion EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 101–105

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
101	**Site/theory**	Garden Watering System | Engineering principles	Teach and unpack this syllabus learning in the Garden Watering System context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Garden Watering System learning module 1** **Open the Garden Watering System project page**	**Identify the types of control systems , including electronic, hydraulic , pneumatic and mechanical** **Official content** **EGT5-MEA-01, EGT5-IVT-01**	Module response showing: Identify the types of control systems , including electronic, hydraulic , pneumatic and mechanical EGT5-MEA-01, EGT5-IVT-01
102	**Project/folio**	Garden Watering System | Engineering principles	Students apply the syllabus learning to their Garden Watering System folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Garden Watering System folio** **Open the Garden Watering System project page**	**Describe how control systems are used to automate processes, improve efficiency and ensure safety in various…** **Official content** **EGT5-SAF-01, EGT5-EVL-01, EGT5-MEA-01**	Folio evidence demonstrating: Describe how control systems are used to automate processes, improve efficiency and ensure safety in various applications EGT5-SAF-01, EGT5-EVL-01, EGT5-MEA-01
103	**Practical**	Garden Watering System | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Garden Watering System making, investigation or testing step and record authentic progress. **Open the Garden Watering System project page**	**Define key terms used to describe the input, control and output principles of a range of control systems** **Official content** **EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Define key terms used to describe the input, control and output principles of a range of control systems EGT5-SAF-01, EGT5-USE-01
104	**Practical**	Garden Watering System | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Garden Watering System making, investigation or testing step and record authentic progress. **Open the Garden Watering System project page**	**Outline regulations, codes of practice, safety protocols and standards that apply in the design and production…** **Official content** **EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Outline regulations, codes of practice, safety protocols and standards that apply in the design and production of an identified control system EGT5-SAF-01, EGT5-USE-01
105	**Practical**	Garden Watering System | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Garden Watering System making, investigation or testing step and record authentic progress. **Open the Garden Watering System project page**	**Recognise the importance of Indigenous Cultural and Intellectual Property (ICIP) of Aboriginal and/or Torres…** **Official content** **EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Recognise the importance of Indigenous Cultural and Intellectual Property (ICIP) of Aboriginal and/or Torres Strait Islander Peoples related to control systems EGT5-SAF-01, EGT5-USE-01

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 106–110

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
106	**Site/theory**	Garden Watering System | Engineering principles	Teach and unpack this syllabus learning in the Garden Watering System context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Garden Watering System learning module 1** **Open the Garden Watering System project page**	**Apply scheduling, resource allocation and budgeting to plan and manage a control system project** **Official content** **EGT5-ENV-01, EGT5-IVT-01**	Module response showing: Apply scheduling, resource allocation and budgeting to plan and manage a control system project EGT5-ENV-01, EGT5-IVT-01
107	**Project/folio**	Garden Watering System | Engineering principles	Students apply the syllabus learning to their Garden Watering System folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Garden Watering System folio** **Open the Garden Watering System project page**	**Identify and apply engineering principles and processes to produce a functioning control system** **Official content** **EGT5-USE-01, EGT5-COM-01**	Folio evidence demonstrating: Identify and apply engineering principles and processes to produce a functioning control system EGT5-USE-01, EGT5-COM-01
108	**Practical**	Garden Watering System | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Garden Watering System making, investigation or testing step and record authentic progress. **Open the Garden Watering System project page**	**Explain the difference between open-loop and closed-loop control systems** **Official content** **EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Explain the difference between open-loop and closed-loop control systems EGT5-SAF-01, EGT5-USE-01
109	**Practical**	Garden Watering System | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Garden Watering System making, investigation or testing step and record authentic progress. **Open the Garden Watering System project page**	**Investigate advanced manufacturing methods to understand their applications, advantages and limitations in…** **Official content** **EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Investigate advanced manufacturing methods to understand their applications, advantages and limitations in producing control systems, such as computer numeric control (CNC) machining, laser, plasma, water jet cutting or rapid prototyping EGT5-SAF-01, EGT5-USE-01

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
110	**Practical**	Garden Watering System | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Garden Watering System making, investigation or testing step and record authentic progress. **Open the Garden Watering System project page**	**Compare renewable and non-renewable resources and explain their advantages and limitations in the design and…** **Official content** **EGT5-ENV-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Compare renewable and non-renewable resources and explain their advantages and limitations in the design and development of control systems EGT5-ENV-01, EGT5-SAF-01, EGT5-USE-01

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 111–115

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
111	**Site/theory**	Garden Watering System | Engineering principles	Teach and unpack this syllabus learning in the Garden Watering System context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Garden Watering System learning module 1** **Open the Garden Watering System project page**	**Investigate innovative applications of a range of control systems, such as artificial intelligence (AI)… Official content** **EGT5-ENV-01, EGT5-IVT-01**	Module response showing: Investigate innovative applications of a range of control systems, such as artificial intelligence (AI), internet of things (IoT) devices or robotics technologies, and how these may affect individuals, society or the environment EGT5-ENV-01, EGT5-IVT-01
112	**Project/folio**	Garden Watering System |	Students apply the syllabus learning to their Garden Watering System folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Garden Watering System folio** **Open the Garden Watering System project page**	**Investigate the use of electronic control systems in everyday situations Official content** **EGT5-COM-01**	Folio evidence demonstrating: Investigate the use of electronic control systems in everyday situations EGT5-COM-01
113	**Practical**	Garden Watering System |	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Garden Watering System making, investigation or testing step and record authentic progress. **Open the Garden Watering System project page**	**Investigate input and output components used in electronic control systems, such as actuators and controllers Official content** **EGT5-USE-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Investigate input and output components used in electronic control systems, such as actuators and controllers EGT5-USE-01, EGT5-SAF-01
114	**Practical**	Garden Watering System |	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Garden Watering System making, investigation or testing step and record authentic progress. **Open the Garden Watering System project page**	**Investigate the use of hydraulic control systems in everyday situations Official content** **EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Investigate the use of hydraulic control systems in everyday situations EGT5-SAF-01, EGT5-USE-01
115	**Practical**	Garden Watering System |	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Garden Watering System making, investigation or testing step and record authentic progress. **Open the Garden Watering System project page**	**Investigate input and output components used in hydraulic control systems, such as reservoir, pump, valve and… Official content** **EGT5-USE-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Investigate input and output components used in hydraulic control systems, such as reservoir, pump, valve and actuators EGT5-USE-01, EGT5-SAF-01

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 116–120

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
116	**Site/theory**	Garden Watering System |	Teach and unpack this syllabus learning in the Garden Watering System context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Garden Watering System learning module 2** **Open the Garden Watering System project page**	**Investigate the use of pneumatic control systems in everyday situations Official content** **EGT5-IVT-01**	Module response showing: Investigate the use of pneumatic control systems in everyday situations EGT5-IVT-01
117	**Project/folio**	Garden Watering System |	Students apply the syllabus learning to their Garden Watering System folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Garden Watering System folio** **Open the Garden Watering System project page**	**Investigate input and output components used in pneumatic control systems, such as a valve, compressor… Official content** **EGT5-USE-01, EGT5-COM-01**	Folio evidence demonstrating: Investigate input and output components used in pneumatic control systems, such as a valve, compressor, regulator and feed line EGT5-USE-01, EGT5-COM-01
118	**Practical**	Garden Watering System |	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Garden Watering System making, investigation or testing step and record authentic progress. **Open the Garden Watering System project page**	**Investigate the use of mechanical control systems in everyday situations Official content** **EGT5-MEA-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Investigate the use of mechanical control systems in everyday situations EGT5-MEA-01, EGT5-SAF-01, EGT5-USE-01
119	**Practical**	Garden Watering System |	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Garden Watering System making, investigation or testing step and record authentic progress. **Open the Garden Watering System project page**	**Investigate input and output components used in mechanical control systems, such as input force , crank and… Official content** **EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Investigate input and output components used in mechanical control systems, such as input force , crank and selector EGT5-USE-01, EGT5-MEA-01, EGT5-SAF-01
120	**Practical**	Garden Watering System | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Garden Watering System making, investigation or testing step and record authentic progress. **Open the Garden Watering System project page**	**Apply safe work practices throughout the design, production and testing of models and projects for a control… Official content** **EGT5-SAF-01, EGT5-EVL-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Apply safe work practices throughout the design, production and testing of models and projects for a control system EGT5-SAF-01, EGT5-EVL-01, EGT5-USE-01

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 121–125

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
121	**Site/theory**	Garden Watering System | Materials	Teach and unpack this syllabus learning in the Garden Watering System context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Garden Watering System learning module 2** **Open the Garden Watering System project page**	**Explore engineering properties of materials suitable for control systems in relation to strength, toughness… Official content** **EGT5-USE-01, EGT5-IVT-01**	Module response showing: Explore engineering properties of materials suitable for control systems in relation to strength, toughness and durability EGT5-USE-01, EGT5-IVT-01
122	**Project/folio**	Garden Watering System | Materials	Students apply the syllabus learning to their Garden Watering System folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Garden Watering System folio** **Open the Garden Watering System project page**	**Test and evaluate the performance of materials used in control systems to determine if they meet required… Official content** **EGT5-SAF-01, EGT5-USE-01, EGT5-EVL-01**	Folio evidence demonstrating: Test and evaluate the performance of materials used in control systems to determine if they meet required specifications and standards for safety, reliability and efficiency EGT5-SAF-01, EGT5-USE-01, EGT5-EVL-01
123	**Practical**	Garden Watering System | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Garden Watering System making, investigation or testing step and record authentic progress. Students complete the teacher-approved collaborative system test and peer verification role. **Open the Garden Watering System project page**	**Evaluate the advantages of the recycling and reuse of materials used in the development of control systems Official content** **EGT5-USE-01, EGT5-EVL-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Evaluate the advantages of the recycling and reuse of materials used in the development of control systems; collaborative role and peer verification record EGT5-USE-01, EGT5-EVL-01, EGT5-SAF-01
124	**Practical**	Garden Watering System | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Garden Watering System making, investigation or testing step and record authentic progress. **Open the Garden Watering System project page**	**Investigate and evaluate the processes used in the recycling of materials, such as metals or polymers Official content** **EGT5-USE-01, EGT5-EVL-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Investigate and evaluate the processes used in the recycling of materials, such as metals or polymers EGT5-USE-01, EGT5-EVL-01, EGT5-SAF-01

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
125	**Practical**	Garden Watering System |	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Garden Watering System making, investigation or testing step and record authentic progress. **Open the Garden Watering System project page**	**Test the electrical properties of materials, such as conductivity and resistance, to identify their… Official content EGT5-USE-01, EGT5-EVL-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Test the electrical properties of materials, such as conductivity and resistance, to identify their application in a control system EGT5-USE-01, EGT5-EVL-01, EGT5-SAF-01

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 126–130

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
126	**Site/theory**	Garden Watering System |	Teach and unpack this syllabus learning in the Garden Watering System context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Garden Watering System learning module 2** **Open the Garden Watering System project page**	**Test the hydraulic properties of materials, such as resistance to pressure and fluids, to identify their… Official content EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01**	Module response showing: Test the hydraulic properties of materials, such as resistance to pressure and fluids, to identify their application in a control system EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01
127	**Project/folio**	Garden Watering System |	Students apply the syllabus learning to their Garden Watering System folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Garden Watering System folio** **Open the Garden Watering System project page**	**Test the pneumatic properties of materials, such as resistance to pressure and fluids, to identify their… Official content EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01**	Folio evidence demonstrating: Test the pneumatic properties of materials, such as resistance to pressure and fluids, to identify their application in a control system EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01
128	**Practical**	Garden Watering System |	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Garden Watering System making, investigation or testing step and record authentic progress. **Open the Garden Watering System project page**	**Test the mechanical properties of materials, such as resistance to wear and fatigue, to identify their… Official content EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01**	Teacher observation and project/test record demonstrating: Test the mechanical properties of materials, such as resistance to wear and fatigue, to identify their application in a control system EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01
129	**Practical**	Garden Watering System | Systems analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Garden Watering System making, investigation or testing step and record authentic progress. **Open the Garden Watering System project page**	**Use an engineering process to produce a functioning control system Official content EGT5-USE-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Use an engineering process to produce a functioning control system EGT5-USE-01, EGT5-SAF-01
130	**Practical**	Garden Watering System | Systems analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Garden Watering System making, investigation or testing step and record authentic progress. **Open the Garden Watering System project page**	**Use collaborative work practices to improve efficiencies in a control system project Official content EGT5-EVL-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Use collaborative work practices to improve efficiencies in a control system project EGT5-EVL-01, EGT5-SAF-01, EGT5-USE-01

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 131–135

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
131	**Site/theory**	Garden Watering System | Systems analysis	Teach and unpack this syllabus learning in the Garden Watering System context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Garden Watering System learning module 3** **Open the Garden Watering System project page**	**Design, construct and test a control system for a specific purpose using appropriate components and… Official content** **EGT5-USE-01, EGT5-EVL-01, EGT5-IVT-01**	Module response showing: Design, construct and test a control system for a specific purpose using appropriate components and considering differences between inputs and outputs EGT5-USE-01, EGT5-EVL-01, EGT5-IVT-01
132	**Project/folio**	Garden Watering System | Systems analysis	Students apply the syllabus learning to their Garden Watering System folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Garden Watering System folio** **Open the Garden Watering System project page**	**Use a range of equipment, hand and power tools, and machines in the construction of projects or working models… Official content** **EGT5-USE-01, EGT5-COM-01**	Folio evidence demonstrating: Use a range of equipment, hand and power tools, and machines in the construction of projects or working models in control systems EGT5-USE-01, EGT5-COM-01
133	**Practical**	Garden Watering System | Systems analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Garden Watering System making, investigation or testing step and record authentic progress. **Open the Garden Watering System project page**	**Test control systems to determine efficiency Official content** **EGT5-EVL-01, EGT5-MEA-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Test control systems to determine efficiency EGT5-EVL-01, EGT5-MEA-01, EGT5-SAF-01
134	**Practical**	Garden Watering System | Systems analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Garden Watering System making, investigation or testing step and record authentic progress. **Open the Garden Watering System project page**	**Assess the integration of motors in control systems to improve efficiency Official content** **EGT5-EVL-01, EGT5-MEA-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Assess the integration of motors in control systems to improve efficiency EGT5-EVL-01, EGT5-MEA-01, EGT5-SAF-01
135	**Practical**	Garden Watering System | Systems analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Garden Watering System making, investigation or testing step and record authentic progress. **Open the Garden Watering System project page**	**Document troubleshooting processes and solutions to maintain a clear record of issues and their resolutions Official content** **EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Document troubleshooting processes and solutions to maintain a clear record of issues and their resolutions EGT5-SAF-01, EGT5-USE-01

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 136–140

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
136	**Site/theory**	Garden Watering System |	Teach and unpack this syllabus learning in the Garden Watering System context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Garden Watering System learning module 3** **Open the Garden Watering System project page**	**Test the fundamental principles of electricity, such as voltage, current and resistance, in series and… Official content** **EGT5-EVL-01, EGT5-IVT-01**	Module response showing: Test the fundamental principles of electricity, such as voltage, current and resistance, in series and parallel circuits, to explain how they are applied in the functioning of electronic control systems EGT5-EVL-01, EGT5-IVT-01
137	**Project/folio**	Garden Watering System |	Students apply the syllabus learning to their Garden Watering System folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Garden Watering System folio** **Open the Garden Watering System project page**	**Use a multimeter to test an electronic circuit to measure voltage and current and to determine resistance Official content** **EGT5-EVL-01, EGT5-COM-01**	Folio evidence demonstrating: Use a multimeter to test an electronic circuit to measure voltage and current and to determine resistance EGT5-EVL-01, EGT5-COM-01
138	**Practical**	Garden Watering System |	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Garden Watering System making, investigation or testing step and record authentic progress. **Open the Garden Watering System project page**	**Test the fundamental principles of hydraulics, such as fluid pressure and flow, to explain how they are… Official content** **EGT5-EVL-01, EGT5-MEA-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Test the fundamental principles of hydraulics, such as fluid pressure and flow, to explain how they are applied in the functioning of hydraulic control systems EGT5-EVL-01, EGT5-MEA-01, EGT5-SAF-01
139	**Practical**	Garden Watering System |	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Garden Watering System making, investigation or testing step and record authentic progress. **Open the Garden Watering System project page**	**Test a hydraulic system using a pressure gauge to assess the integrity of hoses and components Official content** **EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01**	Teacher observation and project/test record demonstrating: Test a hydraulic system using a pressure gauge to assess the integrity of hoses and components EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
140	**Practical**	Garden Watering System |	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Garden Watering System making, investigation or testing step and record authentic progress. **Open the Garden Watering System project page**	**Test the fundamental principles of pneumatics, such as air pressure and volume, to explain how they are… Official content** **EGT5-EVL-01, EGT5-MEA-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Test the fundamental principles of pneumatics, such as air pressure and volume, to explain how they are applied in the functioning of pneumatic control systems EGT5-EVL-01, EGT5-MEA-01, EGT5-SAF-01

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P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
141	**Site/theory**	Garden Watering System |	Teach and unpack this syllabus learning in the Garden Watering System context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Garden Watering System learning module 4** **Open the Garden Watering System project page**	**Test a pneumatic system using a pressure gauge to assess the integrity of hoses and components** **Official content** **EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01**	Module response showing: Test a pneumatic system using a pressure gauge to assess the integrity of hoses and components EGT5-USE-01, EGT5-EVL-01, EGT5-MEA-01
142	**Project/folio**	Garden Watering System |	Students apply the syllabus learning to their Garden Watering System folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Garden Watering System folio** **Open the Garden Watering System project page**	**Test the fundamental principles of mechanics, such as force, motion and energy, to explain how they are…** **Official content** **EGT5-EVL-01, EGT5-MEA-01, EGT5-COM-01**	Folio evidence demonstrating: Test the fundamental principles of mechanics, such as force, motion and energy, to explain how they are applied in the functioning of mechanical control systems EGT5-EVL-01, EGT5-MEA-01, EGT5-COM-01
143	**Practical**	Garden Watering System |	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Garden Watering System making, investigation or testing step and record authentic progress. **Open the Garden Watering System project page**	**Use a block and tackle to test a mechanical system to determine the relationship between load and effort to…** **Official content** **EGT5-EVL-01, EGT5-MEA-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Use a block and tackle to test a mechanical system to determine the relationship between load and effort to explain mechanical advantage EGT5-EVL-01, EGT5-MEA-01, EGT5-SAF-01
144	**Practical**	Garden Watering System | Communication	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Garden Watering System making, investigation or testing step and record authentic progress. **Open the Garden Watering System project page**	**Produce annotated sketches of project components to visualise, communicate, understand and record ideas to…** **Official content** **EGT5-COM-01, EGT5-GRP-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Produce annotated sketches of project components to visualise, communicate, understand and record ideas to develop a control system project EGT5-COM-01, EGT5-GRP-01, EGT5-USE-01
145	**Practical**	Garden Watering System | Communication	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Garden Watering System making, investigation or testing step and record authentic progress. **Open the Garden Watering System project page**	**Develop, read and interpret technical diagrams to prepare materials for the production of a control system…** **Official content** **EGT5-GRP-01, EGT5-USE-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Develop, read and interpret technical diagrams to prepare materials for the production of a control system project EGT5-GRP-01, EGT5-USE-01, EGT5-SAF-01

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 146–150

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
146	**Site/theory**	Garden Watering System | Communication	Teach and unpack this syllabus learning in the Garden Watering System context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Garden Watering System learning module 4** **Open the Garden Watering System project page**	**Modify and apply appropriate engineering drawings in the completion of a control system project Official content EGT5-GRP-01, EGT5-EVL-01, EGT5-IVT-01**	Module response showing: Modify and apply appropriate engineering drawings in the completion of a control system project EGT5-GRP-01, EGT5-EVL-01, EGT5-IVT-01
147	**Project/folio**	Garden Watering System | Communication	Students apply the syllabus learning to their Garden Watering System folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Garden Watering System folio** **Open the Garden Watering System project page**	**Create written texts to explain and evaluate factors that influence the design and engineering of control… Official content EGT5-COM-01, EGT5-EVL-01**	Folio evidence demonstrating: Create written texts to explain and evaluate factors that influence the design and engineering of control systems EGT5-COM-01, EGT5-EVL-01
148	**Practical**	Garden Watering System | Communication	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Garden Watering System making, investigation or testing step and record authentic progress. **Open the Garden Watering System project page**	**Use subject-specific terminology to communicate concepts of control systems Official content EGT5-COM-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Use subject-specific terminology to communicate concepts of control systems EGT5-COM-01, EGT5-SAF-01, EGT5-USE-01
149	**Practical**	Garden Watering System | Communication	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Garden Watering System making, investigation or testing step and record authentic progress. **Open the Garden Watering System project page**	**Explain control system relationships using block diagrams Official content EGT5-GRP-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Explain control system relationships using block diagrams EGT5-GRP-01, EGT5-SAF-01, EGT5-USE-01
150	**Practical**	Garden Watering System | Communication	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Garden Watering System making, investigation or testing step and record authentic progress. **Open the Garden Watering System project page**	**Document material selection and justification, systems analysis and test results in an engineering report when… Official content EGT5-COM-01, EGT5-USE-01, EGT5-EVL-01**	Teacher observation and project/test record demonstrating: Document material selection and justification, systems analysis and test results in an engineering report when developing a control system solution EGT5-COM-01, EGT5-USE-01, EGT5-EVL-01

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 151–155

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
151	**Site/theory**	Wind-powered USB Charger | Engineering principles	Teach and unpack this syllabus learning in the Wind-powered USB Charger context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Wind-powered USB Charger learning module 1** **Open the Wind-powered USB Charger project page**	**Examine the purpose of an identified field of engineering Official content EGT5-IVT-01**	Module response showing: Examine the purpose of an identified field of engineering EGT5-IVT-01
152	**Project/folio**	Wind-powered USB Charger | Engineering principles	Students apply the syllabus learning to their Wind-powered USB Charger folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Wind-powered USB Charger folio** **Open the Wind-powered USB Charger project page**	**Investigate the technical knowledge and skills required in engineering-related industries Official content EGT5-COM-01**	Folio evidence demonstrating: Investigate the technical knowledge and skills required in engineering-related industries EGT5-COM-01
153	**Practical**	Wind-powered USB Charger | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Wind-powered USB Charger making, investigation or testing step and record authentic progress. **Open the Wind-powered USB Charger project page**	Describe the impact of new and emerging technologies on careers and professions in the field of engineering **Official content EGT5-IVT-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Describe the impact of new and emerging technologies on careers and professions in the field of engineering EGT5-IVT-01, EGT5-SAF-01, EGT5-USE-01
154	**Practical**	Wind-powered USB Charger | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Wind-powered USB Charger making, investigation or testing step and record authentic progress. **Open the Wind-powered USB Charger project page**	**Describe the impact of new and emerging technologies on careers and professions in the field of engineering Official content EGT5-IVT-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Describe the impact of new and emerging technologies on careers and professions in the field of engineering EGT5-IVT-01, EGT5-SAF-01, EGT5-USE-01
155	**Practical**	Wind-powered USB Charger | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Wind-powered USB Charger making, investigation or testing step and record authentic progress. **Open the Wind-powered USB Charger project page**	**Outline engineering principles and processes used in an identified field Official content EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Outline engineering principles and processes used in an identified field EGT5-SAF-01, EGT5-USE-01

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 156–160

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
156	**Site/theory**	Wind-powered USB Charger | Engineering principles	Teach and unpack this syllabus learning in the Wind-powered USB Charger context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Wind-powered USB Charger learning module 1** **Open the Wind-powered USB Charger project page**	Investigate and test multiple ways of solving an engineering problem during a project **Official content** **EGT5-EVL-01, EGT5-IVT-01**	Module response showing: Investigate and test multiple ways of solving an engineering problem during a project EGT5-EVL-01, EGT5-IVT-01
157	**Project/folio**	Wind-powered USB Charger | Engineering principles	Students apply the syllabus learning to their Wind-powered USB Charger folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Wind-powered USB Charger folio** **Open the Wind-powered USB Charger project page**	**Develop an engineering brief that includes objectives and constraints for a project in a specialised field Official content** **EGT5-COM-01**	Folio evidence demonstrating: Develop an engineering brief that includes objectives and constraints for a project in a specialised field EGT5-COM-01
158	**Practical**	Wind-powered USB Charger | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Wind-powered USB Charger making, investigation or testing step and record authentic progress. **Open the Wind-powered USB Charger project page**	**Investigate and test multiple ways of solving an engineering problem during a project Official content** **EGT5-EVL-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Investigate and test multiple ways of solving an engineering problem during a project EGT5-EVL-01, EGT5-SAF-01, EGT5-USE-01
159	**Practical**	Wind-powered USB Charger | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Wind-powered USB Charger making, investigation or testing step and record authentic progress. **Open the Wind-powered USB Charger project page**	Test and evaluate to improve projects using performance criteria to develop a solution to an engineering problem **Official content** **EGT5-EVL-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Test and evaluate to improve projects using performance criteria to develop a solution to an engineering problem EGT5-EVL-01, EGT5-SAF-01, EGT5-USE-01
160	**Practical**	Wind-powered USB Charger | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Wind-powered USB Charger making, investigation or testing step and record authentic progress. **Open the Wind-powered USB Charger project page**	**Develop and conduct a risk assessment, including risk mitigation strategies, for an engineering project Official content** **EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Develop and conduct a risk assessment, including risk mitigation strategies, for an engineering project EGT5-SAF-01, EGT5-USE-01

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 161–165

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
161	**Site/theory**	Wind-powered USB Charger | Engineering principles	Teach and unpack this syllabus learning in the Wind-powered USB Charger context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Wind-powered USB Charger learning module 1** **Open the Wind-powered USB Charger project page**	**Develop performance criteria to be used to evaluate issues and select viable solutions during a project Official content EGT5-EVL-01, EGT5-IVT-01**	Module response showing: Develop performance criteria to be used to evaluate issues and select viable solutions during a project EGT5-EVL-01, EGT5-IVT-01
162	**Project/folio**	Wind-powered USB Charger | Engineering principles	Students apply the syllabus learning to their Wind-powered USB Charger folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Wind-powered USB Charger folio** **Open the Wind-powered USB Charger project page**	Assess project goals and outcomes using feedback from stakeholders to inform decisions **Official content EGT5-COM-01**	Folio evidence demonstrating: Assess project goals and outcomes using feedback from stakeholders to inform decisions EGT5-COM-01
163	**Practical**	Wind-powered USB Charger | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Wind-powered USB Charger making, investigation or testing step and record authentic progress. **Open the Wind-powered USB Charger project page**	**Test and evaluate to improve projects using performance criteria to develop a solution to an engineering problem Official content EGT5-EVL-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Test and evaluate to improve projects using performance criteria to develop a solution to an engineering problem EGT5-EVL-01, EGT5-SAF-01, EGT5-USE-01
164	**Practical**	Wind-powered USB Charger | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Wind-powered USB Charger making, investigation or testing step and record authentic progress. **Open the Wind-powered USB Charger project page**	**Evaluate the suitability of a range of energy sources, including the use of sustainable energy for use in an… Official content EGT5-ENV-01, EGT5-EVL-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Evaluate the suitability of a range of energy sources, including the use of sustainable energy for use in an identified project EGT5-ENV-01, EGT5-EVL-01, EGT5-SAF-01
165	**Practical**	Wind-powered USB Charger | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Wind-powered USB Charger making, investigation or testing step and record authentic progress. **Open the Wind-powered USB Charger project page**	Use test results to select suitable materials to solve an engineering problem **Official content EGT5-USE-01, EGT5-EVL-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Use test results to select suitable materials to solve an engineering problem EGT5-USE-01, EGT5-EVL-01, EGT5-SAF-01

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 166–170

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
166	**Site/theory**	Wind-powered USB Charger | Engineering principles	Teach and unpack this syllabus learning in the Wind-powered USB Charger context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Wind-powered USB Charger learning module 2** **Open the Wind-powered USB Charger project page**	**Apply scheduling, resource allocation and budgeting to plan and manage an identified project** Official content **EGT5-ENV-01, EGT5-IVT-01**	Module response showing: Apply scheduling, resource allocation and budgeting to plan and manage an identified project EGT5-ENV-01, EGT5-IVT-01
167	**Project/folio**	Wind-powered USB Charger | Engineering principles	Students apply the syllabus learning to their Wind-powered USB Charger folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Wind-powered USB Charger folio** **Open the Wind-powered USB Charger project page**	**Assess project goals and outcomes using feedback from stakeholders to inform decisions** Official content **EGT5-COM-01**	Folio evidence demonstrating: Assess project goals and outcomes using feedback from stakeholders to inform decisions EGT5-COM-01
168	**Practical**	Wind-powered USB Charger | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Wind-powered USB Charger making, investigation or testing step and record authentic progress. **Open the Wind-powered USB Charger project page**	Investigate advancements in material science and their implications for engineering projects Official content **EGT5-USE-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Investigate advancements in material science and their implications for engineering projects EGT5-USE-01, EGT5-SAF-01
169	**Practical**	Wind-powered USB Charger | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Wind-powered USB Charger making, investigation or testing step and record authentic progress. **Open the Wind-powered USB Charger project page**	**Apply safe work practices throughout the design, production and testing of models and projects for an…** Official content **EGT5-SAF-01, EGT5-EVL-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Apply safe work practices throughout the design, production and testing of models and projects for an identified field EGT5-SAF-01, EGT5-EVL-01, EGT5-USE-01
170	**Practical**	Wind-powered USB Charger | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Wind-powered USB Charger making, investigation or testing step and record authentic progress. **Open the Wind-powered USB Charger project page**	**Research and test potential materials and components suitable for the project** Official content **EGT5-USE-01, EGT5-EVL-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Research and test potential materials and components suitable for the project EGT5-USE-01, EGT5-EVL-01, EGT5-SAF-01

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 171–175

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
171	**Site/theory**	Wind-powered USB Charger | Technical analysis	Teach and unpack this syllabus learning in the Wind-powered USB Charger context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Wind-powered USB Charger learning module 2** **Open the Wind-powered USB Charger project page**	Use software to perform simulations and computational analysis to predict the success of the project **Official content** **EGT5-IVT-01**	Module response showing: Use software to perform simulations and computational analysis to predict the success of the project EGT5-IVT-01
172	**Project/folio**	Wind-powered USB Charger | Materials	Students apply the syllabus learning to their Wind-powered USB Charger folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Wind-powered USB Charger folio** **Open the Wind-powered USB Charger project page**	**Use test results to select suitable materials to solve an engineering problem** **Official content** **EGT5-USE-01, EGT5-EVL-01, EGT5-COM-01**	Folio evidence demonstrating: Use test results to select suitable materials to solve an engineering problem EGT5-USE-01, EGT5-EVL-01, EGT5-COM-01
173	**Practical**	Wind-powered USB Charger | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Wind-powered USB Charger making, investigation or testing step and record authentic progress. **Open the Wind-powered USB Charger project page**	**Investigate the environmental impact and sustainability of different materials to be used in the project** **Official content** **EGT5-ENV-01, EGT5-USE-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Investigate the environmental impact and sustainability of different materials to be used in the project EGT5-ENV-01, EGT5-USE-01, EGT5-SAF-01
174	**Practical**	Wind-powered USB Charger | Technical analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Wind-powered USB Charger making, investigation or testing step and record authentic progress. **Open the Wind-powered USB Charger project page**	Use computer-aided design (CAD) software to develop and modify designs **Official content** **EGT5-GRP-01, EGT5-EVL-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Use computer-aided design (CAD) software to develop and modify designs EGT5-GRP-01, EGT5-EVL-01, EGT5-SAF-01
175	**Practical**	Wind-powered USB Charger | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Wind-powered USB Charger making, investigation or testing step and record authentic progress. **Open the Wind-powered USB Charger project page**	**Compare the cost-effectiveness of various materials for specific applications within the development of the…** **Official content** **EGT5-USE-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Compare the cost-effectiveness of various materials for specific applications within the development of the solution to an engineering problem EGT5-USE-01, EGT5-SAF-01

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 176–180

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
176	**Site/theory**	Wind-powered USB Charger | Materials	Teach and unpack this syllabus learning in the Wind-powered USB Charger context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Wind-powered USB Charger learning module 2** **Open the Wind-powered USB Charger project page**	**Investigate advancements in material science and their implications for engineering projects** **Official content** **EGT5-USE-01, EGT5-IVT-01**	Module response showing: Investigate advancements in material science and their implications for engineering projects EGT5-USE-01, EGT5-IVT-01
177	**Project/folio**	Wind-powered USB Charger | Communication	Students apply the syllabus learning to their Wind-powered USB Charger folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Wind-powered USB Charger folio** **Open the Wind-powered USB Charger project page**	Develop an engineering report using an appropriate format **Official content** **EGT5-COM-01**	Folio evidence demonstrating: Develop an engineering report using an appropriate format EGT5-COM-01
178	**Practical**	Wind-powered USB Charger | Technical analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Wind-powered USB Charger making, investigation or testing step and record authentic progress. **Open the Wind-powered USB Charger project page**	**Develop an engineering drawing for a project using AS 1100** **Official content** **EGT5-GRP-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Develop an engineering drawing for a project using AS 1100 EGT5-GRP-01, EGT5-SAF-01, EGT5-USE-01
179	**Practical**	Wind-powered USB Charger | Technical analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Wind-powered USB Charger making, investigation or testing step and record authentic progress. **Open the Wind-powered USB Charger project page**	**Build a project using a range of tools, machines, equipment and materials** **Official content** **EGT5-USE-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Build a project using a range of tools, machines, equipment and materials EGT5-USE-01, EGT5-SAF-01
180	**Practical**	Wind-powered USB Charger | Communication	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Wind-powered USB Charger making, investigation or testing step and record authentic progress. **Open the Wind-powered USB Charger project page**	Present findings and project outcomes **Official content** **EGT5-COM-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Present findings and project outcomes EGT5-COM-01, EGT5-SAF-01, EGT5-USE-01

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 181–185

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
181	**Site/theory**	Wind-powered USB Charger | Technical analysis	Teach and unpack this syllabus learning in the Wind-powered USB Charger context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Wind-powered USB Charger learning module 3** **Open the Wind-powered USB Charger project page**	**Use software to perform simulations and computational analysis to predict the success of the project** **Official content** **EGT5-IVT-01**	Module response showing: Use software to perform simulations and computational analysis to predict the success of the project EGT5-IVT-01
182	**Project/folio**	Wind-powered USB Charger | Technical analysis	Students apply the syllabus learning to their Wind-powered USB Charger folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Wind-powered USB Charger folio** **Open the Wind-powered USB Charger project page**	**Test components, and record and analyse collected data to assess performance** **Official content** **EGT5-USE-01, EGT5-EVL-01, EGT5-COM-01**	Folio evidence demonstrating: Test components, and record and analyse collected data to assess performance EGT5-USE-01, EGT5-EVL-01, EGT5-COM-01
183	**Practical**	Wind-powered USB Charger | Communication	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Wind-powered USB Charger making, investigation or testing step and record authentic progress. **Open the Wind-powered USB Charger project page**	Use digital tools and platforms for collaborative project management and documentation **Official content** **EGT5-USE-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Use digital tools and platforms for collaborative project management and documentation EGT5-USE-01, EGT5-SAF-01
184	**Practical**	Wind-powered USB Charger | Technical analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Wind-powered USB Charger making, investigation or testing step and record authentic progress. **Open the Wind-powered USB Charger project page**	**Identify issues that inform decision-making in the development of engineering solutions** **Official content** **EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Identify issues that inform decision-making in the development of engineering solutions EGT5-SAF-01, EGT5-USE-01
185	**Practical**	Wind-powered USB Charger | Technical analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Wind-powered USB Charger making, investigation or testing step and record authentic progress. **Open the Wind-powered USB Charger project page**	**Use computer-aided design (CAD) software to develop and modify designs** **Official content** **EGT5-GRP-01, EGT5-EVL-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Use computer-aided design (CAD) software to develop and modify designs EGT5-GRP-01, EGT5-EVL-01, EGT5-SAF-01

Teaching program | Stage 5 Engineering Technology 200-hour Master | Periods 186–190

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
186	**Site/theory**	Wind-powered USB Charger | Engineering principles	Teach and unpack this syllabus learning in the Wind-powered USB Charger context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Wind-powered USB Charger learning module 3** **Open the Wind-powered USB Charger project page**	Describe the impact of new and emerging technologies on careers and professions in the field of engineering **Official content** **EGT5-IVT-01**	Module response showing: Describe the impact of new and emerging technologies on careers and professions in the field of engineering EGT5-IVT-01
187	**Project/folio**	Wind-powered USB Charger | Technical analysis	Students apply the syllabus learning to their Wind-powered USB Charger folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Wind-powered USB Charger folio** **Open the Wind-powered USB Charger project page**	**Implement quality control measures to ensure the accuracy and functionality of the project** **Official content** **EGT5-EVL-01, EGT5-COM-01**	Folio evidence demonstrating: Implement quality control measures to ensure the accuracy and functionality of the project EGT5-EVL-01, EGT5-COM-01
188	**Practical**	Wind-powered USB Charger | Technical analysis	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Wind-powered USB Charger making, investigation or testing step and record authentic progress. **Open the Wind-powered USB Charger project page**	**Investigate ways to incorporate computer-aided manufacturing (CAM) to automate and control manufacturing…** **Official content** **EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Investigate ways to incorporate computer-aided manufacturing (CAM) to automate and control manufacturing processes EGT5-SAF-01, EGT5-USE-01
189	**Practical**	Wind-powered USB Charger | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Wind-powered USB Charger making, investigation or testing step and record authentic progress. **Open the Wind-powered USB Charger project page**	Investigate and test multiple ways of solving an engineering problem during a project **Official content** **EGT5-EVL-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Investigate and test multiple ways of solving an engineering problem during a project EGT5-EVL-01, EGT5-SAF-01, EGT5-USE-01
190	**Practical**	Wind-powered USB Charger | Communication	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Wind-powered USB Charger making, investigation or testing step and record authentic progress. **Open the Wind-powered USB Charger project page**	**Develop an engineering report using an appropriate format** **Official content** **EGT5-COM-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Develop an engineering report using an appropriate format EGT5-COM-01, EGT5-SAF-01, EGT5-USE-01

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P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
191	**Site/theory**	Wind-powered USB Charger | Communication	Teach and unpack this syllabus learning in the Wind-powered USB Charger context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Wind-powered USB Charger learning module 4** **Open the Wind-powered USB Charger project page**	**Create written texts to explain and evaluate processes, challenges and solutions in engineering systems and…** **Official content** **EGT5-COM-01, EGT5-EVL-01, EGT5-IVT-01**	Module response showing: Create written texts to explain and evaluate processes, challenges and solutions in engineering systems and practices EGT5-COM-01, EGT5-EVL-01, EGT5-IVT-01
192	**Project/folio**	Wind-powered USB Charger | Engineering principles	Students apply the syllabus learning to their Wind-powered USB Charger folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Wind-powered USB Charger folio** **Open the Wind-powered USB Charger project page**	Test and evaluate to improve projects using performance criteria to develop a solution to an engineering problem **Official content** **EGT5-EVL-01, EGT5-COM-01**	Folio evidence demonstrating: Test and evaluate to improve projects using performance criteria to develop a solution to an engineering problem EGT5-EVL-01, EGT5-COM-01
193	**Practical**	Wind-powered USB Charger | Communication	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Wind-powered USB Charger making, investigation or testing step and record authentic progress. **Open the Wind-powered USB Charger project page**	**Produce, evaluate and document the completed product** **Official content** **EGT5-USE-01, EGT5-EVL-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Produce, evaluate and document the completed product EGT5-USE-01, EGT5-EVL-01, EGT5-SAF-01
194	**Practical**	Wind-powered USB Charger | Communication	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Wind-powered USB Charger making, investigation or testing step and record authentic progress. **Open the Wind-powered USB Charger project page**	**Present findings and project outcomes** **Official content** **EGT5-COM-01, EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Present findings and project outcomes EGT5-COM-01, EGT5-SAF-01, EGT5-USE-01
195	**Practical**	Wind-powered USB Charger | Engineering principles	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Wind-powered USB Charger making, investigation or testing step and record authentic progress. **Open the Wind-powered USB Charger project page**	Assess project goals and outcomes using feedback from stakeholders to inform decisions **Official content** **EGT5-SAF-01, EGT5-USE-01**	Teacher observation and project/test record demonstrating: Assess project goals and outcomes using feedback from stakeholders to inform decisions EGT5-SAF-01, EGT5-USE-01

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P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
196	**Site/theory**	Wind-powered USB Charger | Communication	Teach and unpack this syllabus learning in the Wind-powered USB Charger context. Students complete the linked module prompt and record a concise explanation or calculation. **Open Wind-powered USB Charger learning module 4** **Open the Wind-powered USB Charger project page**	**Communicate using appropriate data visualisation techniques to support reports and presentations** **Official content** **EGT5-COM-01, EGT5-IVT-01**	Module response showing: Communicate using appropriate data visualisation techniques to support reports and presentations EGT5-COM-01, EGT5-IVT-01
197	**Project/folio**	Wind-powered USB Charger | Communication	Students apply the syllabus learning to their Wind-powered USB Charger folio through labelled graphics, reasoned decisions, calculations, test evidence or evaluation as appropriate. **Open the Wind-powered USB Charger folio** **Open the Wind-powered USB Charger project page**	**Use communication skills to engage in peer review to incorporate feedback and improve project outcomes** **Official content** **EGT5-COM-01, EGT5-EVL-01**	Folio evidence demonstrating: Use communication skills to engage in peer review to incorporate feedback and improve project outcomes EGT5-COM-01, EGT5-EVL-01
198	**Practical**	Wind-powered USB Charger | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Wind-powered USB Charger making, investigation or testing step and record authentic progress. **Open the Wind-powered USB Charger project page**	Use test results to select suitable materials to solve an engineering problem **Official content** **EGT5-USE-01, EGT5-EVL-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Use test results to select suitable materials to solve an engineering problem EGT5-USE-01, EGT5-EVL-01, EGT5-SAF-01
199	**Practical**	Wind-powered USB Charger | Communication	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Wind-powered USB Charger making, investigation or testing step and record authentic progress. **Open the Wind-powered USB Charger project page**	**Use digital tools and platforms for collaborative project management and documentation** **Official content** **EGT5-USE-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Use digital tools and platforms for collaborative project management and documentation EGT5-USE-01, EGT5-SAF-01
200	**Practical**	Wind-powered USB Charger | Materials	Following the current teacher demonstration and local risk controls, students apply this learning during the next approved Wind-powered USB Charger making, investigation or testing step and record authentic progress. **Open the Wind-powered USB Charger project page**	Compare the cost-effectiveness of various materials for specific applications within the development of the… **Official content** **EGT5-USE-01, EGT5-SAF-01**	Teacher observation and project/test record demonstrating: Compare the cost-effectiveness of various materials for specific applications within the development of the solution to an engineering problem EGT5-USE-01, EGT5-SAF-01